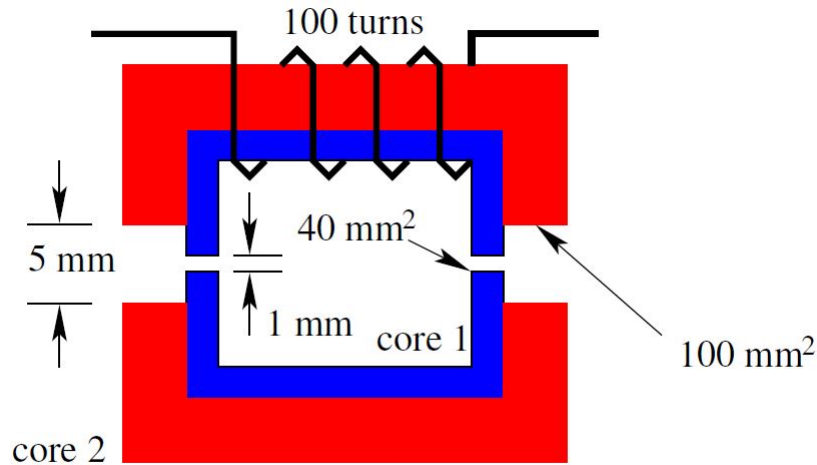
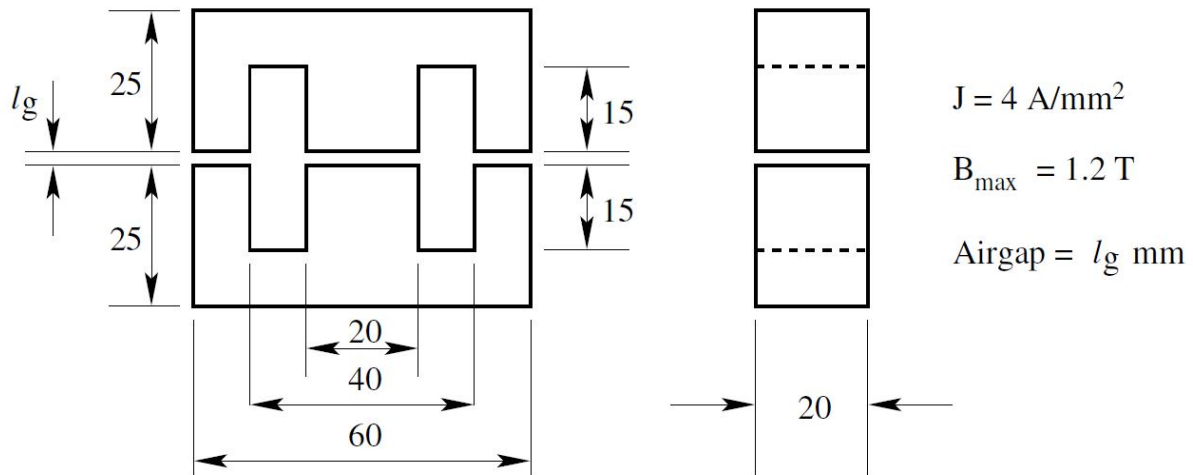


Assignment 5: Magnetics, Inductors and Transformers.

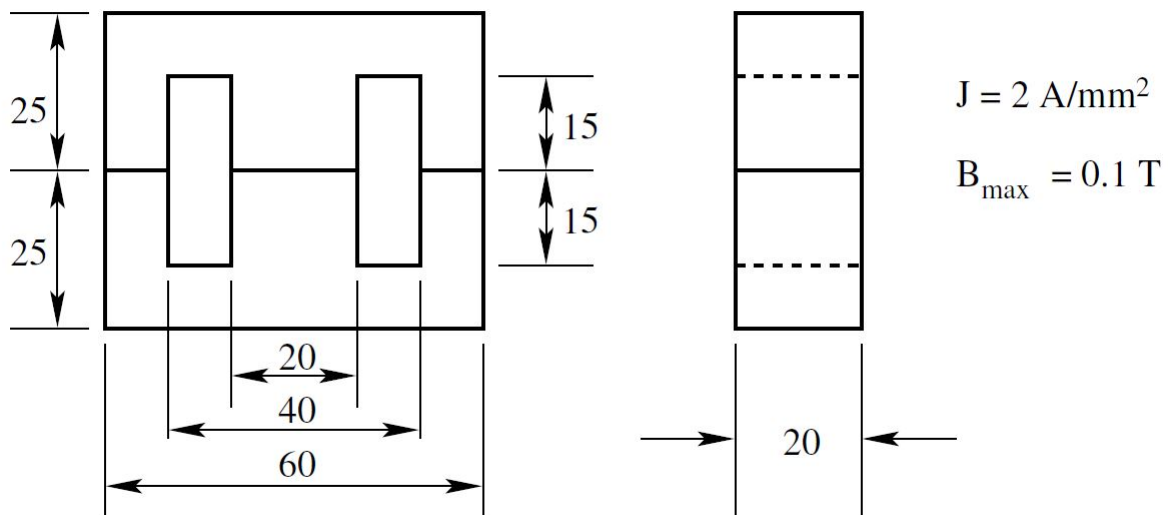
1. The figure shows a magnetic circuit with 100 turns of winding. The magnetic circuit consists of two separate cores (core 1 with 40 mm^2 and core 2 with 100 mm^2). Assume the magnetic materials to be perfect ($\mu_r = \infty$). The core 1 and core 2 saturate at $B_m = 0.2 \text{ T}$. Assume the saturation to be perfect. Make suitable assumptions, if required. (5)



- A. Evaluate the inductance on account of core 1 alone. Also, evaluate its saturation current.
 - B. Evaluate the inductance on account of core 2 alone. Also, evaluate its saturation current.
 - C. Sketch the $\lambda - I$ characteristics of the composite core. λ is the flux linkage.
2. The figure shows the core of a magnetic circuit used for building an inductor for dc application carrying a current of 1 A dc. The core consists of a set of E shaped halves. The dimensions are in mm . The core material has a saturation flux density of 1.2 T . Copper conductors used for this inductor has a maximum allowable current density of 4 A/mm^2 . Take window factor $K_w = 0.5$. Make suitable assumptions, if required. (5)
- A. Evaluate the core area A_c for this core.
 - B. Evaluate the window area A_w for this core.
 - C. Evaluate the maximum energy that can be stored in this core as an inductor.
 - D. Evaluate the largest value of inductance that can be designed with this core for this application.
 - E. Evaluate the number of turns and l_g in mm corresponding to the largest inductor.
 - F. Comment on the effect of fringing in this design.



3. The figure shows the core of a magnetic circuit used for building a transformer for a high frequency application. The excitation voltage for the transformer is a square wave, 400 V peak-to-peak, 25 kHz frequency. The desired secondary voltage is 32 V peak-to-peak. The core consists of a set of E shaped halves. The dimensions are in *mm*. The core material has a saturation flux density of 0.1 T. Copper conductors used for this transformer have a maximum allowable current density of 2 A/mm². Take window factor $K_w = 0.5$. Make suitable assumptions, if required. (5)



- Evaluate the maximum VA rating of the transformer that can be designed with this core for this application.
- Sketch the periodic core flux as a function of time for maximum VA.

- C. Evaluate the number of turns for the primary and secondary windings.
- D. Considering that the core relative permeability is 4000, evaluate the primary magnetising inductance of the transformer.
- E. Sketch the steady state primary magnetising current as a function of time and mark the salient features.